

MINAKSHI THAPA

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EDUCATION

University of Pittsburgh, School of Computer and Information Science Bachelor of Science in Computer Science	Expected: Dec 2027
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SKILLS

Programming Language: Java, Python, C++, JavaScript, SQL, HTML, CSS
Framework/Libraries: Spring Boot, React.js, Node.js, Angular.js, Hibernate, Flutter, .NET MVC, Selenium, Pandas, NumPy
Databases/Tools: MySQL, OracleDB, MongoDB, MS SQL Server, PostgreSQL, Git, GitLab, CI/CD, AWS, Firebase, Docker
AI/ML: NLP, ML Supervised/Unsupervised Learning, Model Evaluation, Hugging Face, TensorFlow, Scikit-learn, PyTorch

EXPERIENCES

AI/ML Fellowship, Break Through Tech	Aug 2025 – Present
- Building RoBERTa NLP classification model using Python achieving 91% F1 score on 165K Reddit banking posts, improving 24% points over TF-IDF baseline using transfer learning and focal loss - Engineering FastAPI backend to support AI-driven chat and analytics, integrating pgvector semantic retrieval with OpenAI-powered LLM responses via REST APIs using AWS S3 for data storage - Deploying Flutter-based web application to Vercel to generate real-time customer sentiment analysis and chat querying, measured by sub-second UI updates during live model inference	

Software Engineering Intern, Bank of New York Mellon	Jun 2025 – Aug 2025
- Engineered REST APIs for Incident Risk Surveillance system using Java Spring, and Oracle DB, cutting API latency by 40%, supporting platform handling 10M+ daily risk evaluations within BNY's global client infrastructure - Developed automated UI testing framework using Java and Selenium integrated into GitLab CI/CD pipelines, reducing regression testing cycles by 33%	

Above and Beyond Computer Science Fellow, Meta	Jan 2024 – Mar 2024
- Mastered data structures and algorithms through technical training program leading to the implementation of efficient solutions for complex programming challenges - Engaged in weekly 25+ code reviews with professional mentors, resulting in development of effective problem-solving strategies and refinement of technical skills	

Software Engineering Intern, Develop for Good	May 2024 – Aug 2024
- Redesigned UI for educational platform using React.js, improving mobile responsiveness and accessibility by 40%, leading to a 30% increase in user engagement - Optimized RESTful APIs using Node.js managing upwards of 15,000 daily user transactions, resulting in latency reduction by over 32% and improving request throughput	

PROJECTS

Income Prediction Machine Learning Model	July 2025
- Built a full ML pipeline in Python on the U.S. Census Adult dataset, achieving 92% ROC-AUC by performing EDA, feature engineering, and training in Random Forest and Logistic Regression models - Evaluated model performance through hyperparameter tuning, ROC/F1/recall metrics, and feature-importance analysis to identify key predictors and enhance generalization	
AutoComplete Engine	Jan 2024
- Developed a Java autocomplete engine with a custom DLB Trie, optimizing search speed and memory usage for predictive text input, addressing slow input speeds - Designed a modular Java API with dynamic word suggestions and user-history tracking to improve prediction relevance and integration flexibility	
Full Stack Spring Boot	Oct 2023
Built a server-side full-stack web application using Java Spring Boot MVC on REST controllers, Hibernate/JPA, to centralize authenticated administrative workflows and file uploads across employee, department, user, and gallery domains.	

LEADERSHIP & EXTRACURRICULAR

Director of Initiatives, Pitt Computer Science Club PITTCSC GITHUB	May 2023 – Present
Managed PittCSC's internship repository and largest hackathon at Pitt, improving transparency for students across United States and delivering 65+ projects with 200+ participants respectively	
Corporate Relations Chair, Pitt Society of Women Engineers	May 2023 – Present
Led 13+ corporate partnerships with companies like Microsoft, J&J, securing \$12,000+ funding, increasing sponsorship and event participation by 40%	
Harvard Women Engineers Code, WeCode, Tech Fellow	Nov 2023 – Mar 2024
Selected as 1 of 65 global fellows, contributing to organizing events and workshops that engaged 955+ attendees from 300+ universities to support women in engineering	